NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

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Abstract

Nested-Space Cosmology (NSC) investigates whether one inherited geometric law can connect nested regions, their exchange of energy across event horizons, and the quantities measured by an interior observer. The foundation rests on five openly declared geometric axioms (**A0--A4**): (**A0**) geometry follows the path of least resistance by minimizing the 1D Cauchy--Schwarz generator stretch cost plus enclosing compression work, $E = Q_{stretch} + PV = \sum_{i=1}^{3} m_i/\lambda_i + PV_0\lambda_1\lambda_2\lambda_3$, which arises directly as the 1D curve-bundle and cavity-mode reduction of the covariant 3+1D field Hamiltonian $\int_V (T^{00}_{phase} + V(s_*))\, d^3x$ with background vacuum pressure $P = V(s_*)$; (**A1**) a 1D spatial seed plus an inherited compression offset (time) generates a closed 2D circle and 3D sphere; (**A2**) seen edge-on, the closed circle $z(\tau)=R_0 e^{i\omega\tau}$ is the harmonic pendulum $R(\tau)=R_0\cos(\omega\tau)$, which crosses `0` smoothly into the negative direction $R<0$ (`-`), flipping the 3D orientation Jacobian $\det F=(R/R_0)^3$ from `+` to `-` and turning inward implosion ($\dot{R}<0$) into outward child expansion ($\dot{r}_- = -\dot{R} > 0$) without the artificial velocity cusp $\Delta\dot{r} = +2R_0\omega$ created when radius is restricted to an unsigned size $r=|R|\geq 0$, while in 3+1D curved geometry the standing-wave hard core $R_{core}=\sqrt{2\pi/\kappa}$ and hybrid radial inertia $M(r)=1/(1+r^2/3)$ bound all curvature invariants across the transition; (**A3**) each nested room carries a moving wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its Planck cutoff (λ_{min}) and Absolute-Zero horizon (λ_{max}); and (**A4**) every room sits between an outer birth `0` valve and inner black-hole `0` valves on a closed vortex--diffusion ring (`0 <-> 0`).

Minimizing the unified 5-variable single-cost functional across `+`, `0`, and `-` yields a unique closed-form global minimum (Section 9) and six parameter-free geometric and dynamical theorems:

- **(1) Exact algebraic budget surface and Friedmann equations of state:** Eliminating both internal variables (θ^*,χ) from the five-variable budget shares (D,M,V) with $B=M-D/3$ yields the exact zero-parameter algebraic curve $9DVB=4(V+B/2)^3$, exact inverse recovery $\theta^*=D/(D+V+B/2)$ and $e^{-\chi}=3\sqrt{\theta^*}B/(2V+B)$ (equivalently $e^{2\chi}=V(D+V+B/2)/[B(V+B/2)]$), and the deep-nest invariant $\lim_{\chi\to\infty} B/V^2=16/27$. Differentiating with respect to the expanding child volume $V_-=V_0(\theta^*)^{3/2}=V_0\, a_-^3$ yields the exact Friedmann equations of state $w_{DE}=-(\partial E_{DE}^*/\partial V_-)/\rho_{DE}=-1$ ($\rho_{DE}=3P=const$) and $w_{DM}=-(\partial E_{DM}^*/\partial V_{bulk})/\rho_{DM}=0$ ($\rho_{DM}\propto a_-^{-3}$, $c_{s,DM}^2=0$), matching the covariant massive-scalar late tail $(p_\chi)/(\rho_\chi)\to 0$.
- **(2) Universal exponential-wall bypasses, `(1,1,1)` 3D triad ground state, and $O(1)$ dilution survival:** Above $\chi_{spaghetti}=(1)/(2)\ln 2$, the infalling `+` parcel shrinks in volume while spaghettifying into a 1D filament ($\theta^*=1/2$), while three scale-free geometric mechanisms ($d\lambda/\lambda$ logarithmic measure, active vortex-tube circulation $w_{tube}=1$, and zero-mode topological phase $\Delta\theta/\pi=1/2$) bypass the e^{-3K} volumetric dilution wall at $O(1)$. In 3D standing-wave closure, the orthogonal `(1,1,1)` triad is the unique minimum-cost 3D ground state ($E_{1,1,1}=3\pi^2/R^2$, 2:1 split against the S^2 angular floor) with wave-compression hard core $R_{core}=\sqrt{2\pi/\kappa}$ and short-range nuclear hard-core repulsion $\Delta E_{core}=8(\sqrt{2}-1)$ against merging into `(2,2,2)`.
- **(3) Surrounding-flow 3:1 Virial partition, helical buckling, and damped `+/0/-` oscillator:** For any incompressible irrotational potential flow around a spherical throat of radius $R(\tau)$ with volume flux $Q=4\pi R^2\dot{R}$, the exterior kinetic energy $T_{ext}=2\pi\rho_0 R^3\dot{R}^2$ and surrounding-flow Virial work $W_{flow}=\int_{r>R} x\cdot(-\nabla p)\, d^3x$ obey the exact parameter-free 3:1 partition $T_{ext}:W_{flow}=3:1$, while axial compression buckles laterally into a helix and drives a damped `+/0/-` horizon oscillation with exact asymmetries $\eta_{area}=\tanh((\pi\gamma)/(2\omega))$ and $\eta_{work}=\tanh((\pi\gamma)/(\omega))=2\eta_{area}/(1+\eta_{area}^2)$.
- **(4) Coupled finite-parent / receiver continuation, continuum zero-radius opening, and last-echo cutoff:** Coupling a finite parent reservoir $E_p(\tau)+E_-(\tau)=E_{tot}$ with hybrid radial inertia $M(r)=1/(1+r^2/3)$ admits a continuum wave-driven opening from zero radius $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$, the closed 3:1 relaxation law $R(\tau)/R_\infty=\tanh(\sqrt{2}E_0\,\tilde{\tau})$, and a sharp last-echo horizon cutoff ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot 25.63%, parent curvature backreaction shift 8.98%).

- **(5) Spherical fragmentation, whole-exterior causal neck onset, and outgoing multipole waveform:** Integrating the outward causal stress balance over the entire exterior $r\in[R,\infty)$ of the exact harmonic two-center family gives the parameter-free characteristic cubic $p^3-3p-3=0$ and static neck-instability root $p_{crit}=1+2^{1/3}+2^{-1/3}$, finite-time neck pinch, positive central observer pulse, and an analytical pole-sum late-time tail $\varphi_C(T)\propto -1/T'+O((T')^{-2})$.
- **(6) Three-axis orthogonal post-pinch cascade and 2×2 kinetic handoff:** Harmonicity ($\Delta\Phi=0$) forces the directional enclosed-volume derivative trace $\partial_{\zeta_x} v+\partial_{\zeta_y} v+\partial_{\zeta_z} v=0$ to vanish at every stage, driving a 3-axis orthogonal cascade $e_z\to e_x\to e_y$ ($1\to 2\to 4\to 8$ lobes with exact harmonic ratios $-1/2$ and -2), coupled by the exact multipole identity $\partial_{q_2} g_{l,0}|_{q_2=0}=-(l-1)/(2)\partial_{q_1} g_{l,0}$ and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix ($\det M_{full}>0$, initial transverse conjugate momentum $p_2(0)>0$).

Sections 2--8 present the covariant 3+1D two-scalar Einstein control apparatus and the pre-registered single-horizon flow test NSC-P006 (*88.47%* scalar patch-energy crossing, normalized area growth $1.0856\ \varepsilon^2$, short-leg transfer $2.2198\ \varepsilon^2$, and one-mode neck-trapping threshold $\varepsilon_{crit}=0.8339$). Secondary single-valve epoch and ladder ansätze ($\Omega_{vis}(\chi_*)=e^{-3}$ and $K=7$) are strictly quarantined in Section 9.7 from the parameter-free theorems, while Appendix A summarizes the nine technical engines and provides the complete reproducibility index.

1. One spatial seed, offset history and a generated geometry

Douglas's proposal is that contrast and exchange organize approximately spherical regions, and that the construction can repeat within them. Time names the offset operation in this generating picture. A physical duration is a readout along the resulting history. The generated space has three spatial directions; a child is another region, not a fourth primitive spatial direction.

The circle and sphere separate an inside from an outside. Isotropic positive surface energy at fixed volume selects a sphere by the isoperimetric inequality. Whether a particular gravitational region is selected by that variational problem must follow from its action. The present smooth horizon has no material tension. A material-shell branch is therefore retained separately in the technical atlas.

The explicit two-stage offset generator is

$$\theta_{n+1}=\theta_n+\alpha$$
$$\phi_{n+1}=\phi_n+\theta_n$$

with phases taken modulo one and zero initial phases ($\theta_0=\varphi_0=0$). Hence $\theta_n=n\alpha$ and $\varphi_n=(1)/(2)n(n-1)\alpha$ modulo one. One accumulator feeds the second; no independent second phase is supplied. For irrational α, Weyl's polynomial equidistribution criterion proves that the joint phase history is dense on the two-phase torus T^2.

Setting $\Theta_n=2\pi\theta_n$ and $\Phi_n=2\pi\varphi_n$, the generated unit directions and their inner-product kernel are

$$v_n=(\sin\Theta_n\cos\Phi_n,\sin\Theta_n\sin\Phi_n,\cos\Theta_n)$$
$$C_{nm}=v_n\cdot v_m.$$

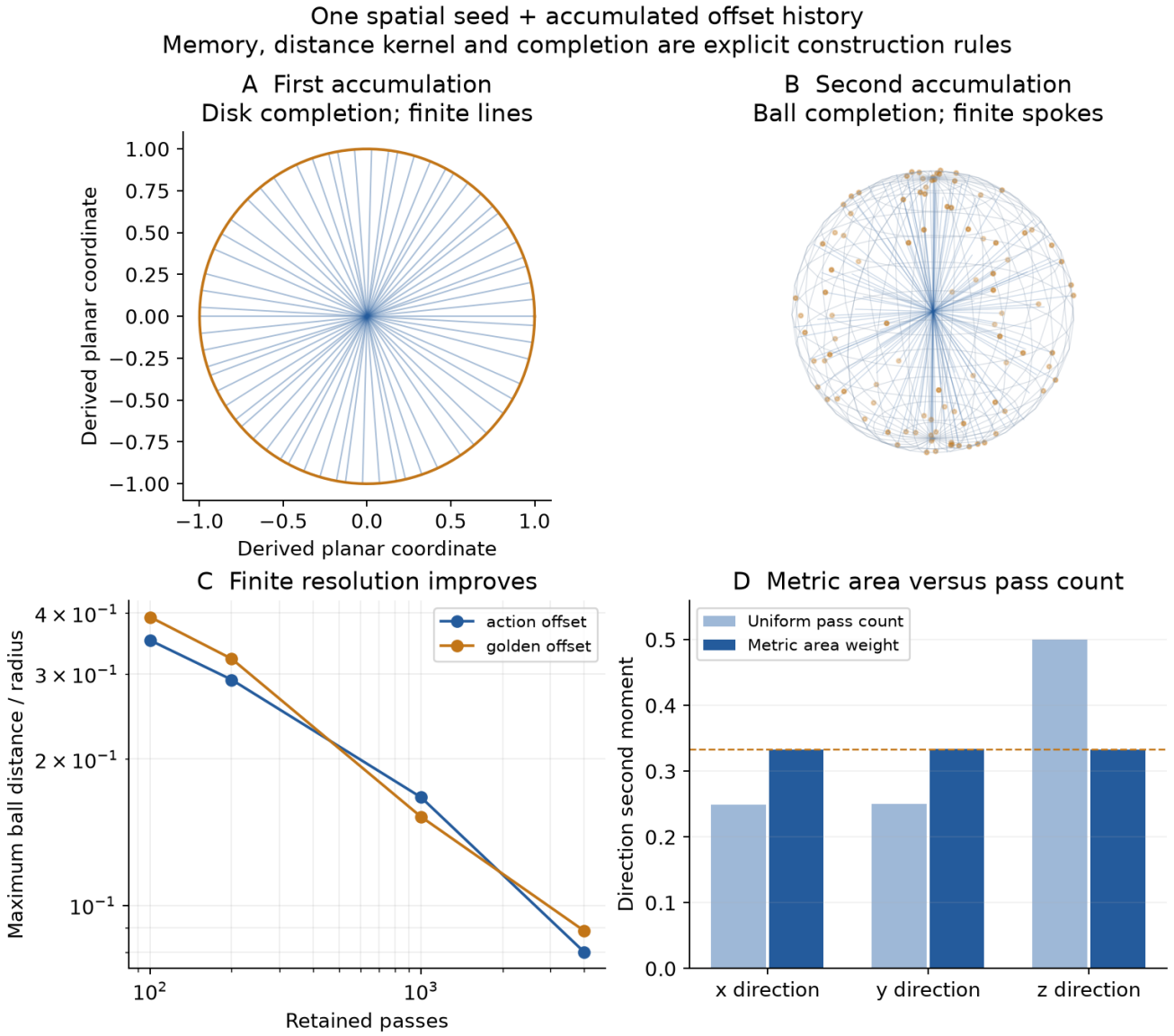
For a retained radial interval, the declared Euclidean distance kernel

$$d^2[(r,n),(s,m)]=r^2+s^2-2rsC_{nm}$$

has the closed three-dimensional ball B^3 as its metric completion. The positive-curvature version

$$d = R\arccos\left[\cos(r/R)\cos(s/R) + \sin(r/R)\sin(s/R)C_{nm}\right]$$

generates the closed spatial parent S^3. Memory, the distance law and metric completion are explicit assumptions; finite drawings have zero three-dimensional volume before completion.



The inherited-law target across nested transitions is

$$\mathcal{T}^{*}\,\mathbb{D} = \mathbb{D}$$

stating that the normalized operator law is preserved under the parent-to-child pullback T^*.

1.1 Openly declared geometric axioms (A0--A4) and covariant field reduction

To build a rigorous deductive chain without unstated fluid-physics analogies (such as assuming parcel incompressibility $\lambda_1\lambda_2\lambda_3=1$ or borrowed surface tension), the non-local foundation states five explicit geometric axioms:

- **A0 (Path of Least Resistance):** By Cauchy--Schwarz on any generated 1D phase curve of physical length $L_i=\lambda_i L_{i,0}$ carrying winding change $\Delta\psi_i$, the intrinsic stretch cost is $Q_{stretch}(A)=(1)/(2)Tr[(A^TA)^{-1/2}C^TC]=\sum_{i=1}^{3} m_i/\lambda_i$. Under enclosing compression work PV, geometry minimizes the single cost functional $E=Q_{stretch}+PV$.

- **A1 (1D Seed + Inherited Offset = Time):** Every child seed forms inside an already evolving parent at relative log-length compression $\chi_{p\to c}=(1)/(3)ln[(J_p/J_{p,0})/(J_c/J_{c,0})]$, which satisfies exact telescoping $\chi_{g\to c}=\chi_{g\to p}+\chi_{p\to c}$ across generations. Repeated offset accumulation generates the 2D circle and 3D sphere.
- **A2 (Signed-Zero Crossing + → 0 → -):** The radial coordinate $R(\tau)$ is a signed geometric coordinate rather than an unsigned size. Under winning inward compression $F_{net}=F_{in}-F_{out}>0$, $R(\tau)$ crosses smoothly through `0` into $R<0$ (`-`), flipping the 3D orientation Jacobian $det[(R/R_0)I_3]=(R/R_0)^3$ from `+` to `-` and converting inward implosion ($R<0$) into outward child expansion ($\dot{r}_-=-R>0$, $r_-=-R>0$).
- **A3 (Moving Spectral Band $[\lambda_{min},\lambda_{max}]$):** Each room at relative compression χ supports a finite wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its shortest wavelength before `0` crosses into `-` (local Planck cutoff λ_{min}) and its longest wavelength as wide as the room itself (local Absolute-Zero horizon $\lambda_{max}=2R$).
- **A4 (Two-Valve Nested Circulation):** Each `0` horizon is a one-way valve from `+` to `-` conserving total turn budget $n_++n_-=N_0$ ($n_+=(1-\theta)N_0$, $n_-=\theta N_0$). Every local room sits between an outer birth `0` valve (inflow from parent) and inner black-hole `0` valves (outflow to children) on a closed vortex--diffusion ring (`0 <-> 0`).

Proposition 1.1 (Reduction of the Covariant 3+1D Field Hamiltonian to A0): Consider the covariant 3+1D scalar-gravity Lagrangian of Section 2, $L=(R)/(2\kappa)-(1)/(2)(\nabla S)^2-(1)/(2)(\nabla\chi)^2-V(s)-(1)/(2)\mu^2(s)\chi^2$, evaluated on an orthorhombic principal domain $V=V_0\lambda_1\lambda_2\lambda_3$ of physical lengths $L_i=\lambda_i L_{i,0}$ ($i=1,2,3$) immersed in the parent background vacuum energy density $P\equiv V(s_*)$. Along each generated 1D phase-curve bundle γ_i of physical length $L_i=\lambda_i L_{i,0}$ and conserved cross-sectional flux tube $A_{\perp,i}$ carrying topological phase winding $\Delta\psi_i=\int_0^{L_i}\partial_s\psi_i\,ds=2\pi n_i$ (or, equivalently, for a relativistic cavity standing wave with principal wavevector $k_i=\pi n_i/(\lambda_i L_{i,0})$), the Cauchy--Schwarz inequality on the spatial gradient energy yields

$$\int_{\text{tube}_i}\frac{1}{2}(\partial_s\psi_i)^2\,d^3x=\frac{A_{\perp,i}}{2}\int_0^{L_i}(\partial_s\psi_i)^2\,ds\geq\frac{A_{\perp,i}(\Delta\psi_i)^2}{2L_{i,0}\lambda_i}\equiv\frac{m_i}{\lambda_i}$$

$$m_i=\frac{2\pi^2A_{\perp,i}n_i^2}{L_{i,0}}$$

with equality attained on the harmonic uniform-gradient phase $\partial_s\psi_i=\Delta\psi_i/L_i$. Adding the bulk background vacuum energy $\int_V V(s_*)\,d^3x=PV_0\lambda_1\lambda_2\lambda_3$ gives the exact reduced Hamiltonian functional of **A0**:

$$E(\lambda_1,\lambda_2,\lambda_3)=\int_V\left(T^{00}_{\text{phase}}+V(s_*)\right)d^3x=\sum_{i=1}^{3}\frac{m_i}{\lambda_i}+PV_0\lambda_1\lambda_2\lambda_3.$$

Whereas smearing a fixed gradient homogeneously across empty 3D volume dilutes as $\lambda^{d-2}=\lambda^1=e^{-\chi}$, integrating along the generated 1D phase curves preserves the $1/\lambda_i$ Cauchy--Schwarz stretch resistance in every dimension d.

1.2 Circle-pendulum duality: why `-` exists, `(-) x (-) = (+)`, and curvature regularization at $R=0$

In Layer 1 of the offset generator, the pen draws a closed circle $z(\tau)=R_0e^{i\omega\tau}$. Seen edge-on along any diameter, that uniform circular motion is the harmonic pendulum

$$R(\tau)=\text{Re}\left(R_0e^{i\omega\tau}\right)=R_0\cos(\omega\tau)$$

$$\ddot{R}+\omega^2R=0$$

which swings $+R_0 \to 0 \to -R_0 \to 0 \to +R_0$ with invariant total energy $(1)/(2)R^2+(1)/(2)\omega^2\ R^2=(1)/(2)R_0{}^2\omega^2$. Three exact geometric consequences follow immediately:

- **(1) Why - must exist for space to close:** Negative is direction, not less than nothing. If negative coordinates are forbidden ($R \geq 0$ only), the circle is clipped to $\omega\tau \in [-\pi/2, \pi/2]$: exactly *50%* of its arc length (πR_0 out of $2\pi R_0$) is lost and no loop or bounded space can close.
- **(2) Why `(-) x (-) = (+)` is two-generation geometry:** Multiplying by $-1=e^{i\pi}$ is a geometric half-turn (π rotation). In three spatial dimensions, crossing `0` applies the spatial inversion $P=-I_3$ with $\det P=(-1)^3=-1$ (*Parent (+)→Child (-)*). Crossing `0` a second time into the grandchild applies two half-turns, $P^2=(-I_3)(-I_3)=+I_3$ with $\det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*).
- **(3) Kinematic cusp removal and 3+1D curvature regularization at $R=0$:** Standard physics uses negative coordinates everywhere except for the radius of space itself, which it restricts to an unsigned size $r(\tau)=|R(\tau)| \geq 0$. At the horizon crossing $\tau_0=\pi/(2\omega)$ ($R=0$), folding the negative half of the swing onto the positive half forces a discontinuous velocity jump $\Delta\dot{r}=\dot{r}(\tau_0^+)-\dot{r}(\tau_0^-)=+2R_0\omega$ and a Dirac-delta acceleration cusp $\ddot{r}=2R_0\omega\,\delta(\tau-\tau_0)-\omega^2 r$. Keeping the signed coordinate $R(\tau)$ removes this kinematic cusp completely ($\Delta\dot{R}=0$, $\ddot{R}(\tau_0)=0$). In 3+1D curved geometry, where coordinate-invariant curvature scalars such as the Kretschmann invariant $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \propto R_{areal}^{-6}$ would diverge if a massive shell collapsed to zero physical area, this signed coordinate passage acts in tandem with two physical regularizations derived in Section 9: the `(1,1,1)` standing-wave compression hard core $R_{core}=\sqrt{2\pi/\kappa}$ (Theorem 2, Section 9.2), which bounds spatial curvature invariants by $R_{areal} \geq R_{core}>0$, and the hybrid radial inertia metric $M(r)=1/(1+r^2/3)$ (Theorem 4, Section 9.4), which makes the proper birth distance $\int_0^{\varepsilon}\sqrt{M(r)}\,dr<\infty$ finite and admits a smooth continuum opening from zero radius.

2. The smooth zero interface and its inherited source

Before presenting the complete **A0--A4** variational theorems in Section 9, Sections 2--8 record the conditional two-scalar Einstein control apparatus. This control apparatus uses the orientation convention **+ source/energy, 0 smooth horizon, - receiving/drain** to test what an isolated single-horizon scalar flow can and cannot determine on its own.

The worked parent uses the Einstein/two-scalar action

$$\mathcal{L} = \frac{R}{2\kappa} - \frac{(\nabla S)^2}{2} - \frac{(\nabla \chi)^2}{2} - W$$
$$W = V(s) + \frac{\mu^2(s)\chi^2}{2}$$

with $S=\sqrt{84}(s-s_*)$ and

$$V(s) = 7e^{-12s} - 7.5e^{-14s} + \frac{25}{32}e^{-24s}$$
$$\mu^2(s) = 400e^{-12s}.$$

Its prepared real scalar contains two spatial harmonics with Laplacian eigenvalues *3* and *8*, equal reference energy, and zero initial velocity. The reference frequency $\omega_0=20$, state amplitude ε, and background curvature H remain declared inputs; the minimum of $V(s)$ supplies the background parent vacuum $P=V(s_*)$.

In conformal null coordinates $u=\eta-\chi$ and $v=\eta+\chi$, the smooth horizon lies at $v=\pi/2$, where the inherited radial fields satisfy

$$4\Psi_{uv} + \left[(m^2 - 2)\sec^2\eta + 2\csc^2\chi\right]\Psi = 0$$
$$m^2 = \frac{\mu^2}{H^2}.$$

There is no horizon denominator and no membrane stress. Two intersecting null traces inherited from the global preparation fix the receiving field across $v=\pi/2$ without resetting the state.

3. One source, three connected crossing numbers

The source's null stress gives a positive regional Killing-energy flux:

$$\frac{dE_K}{dt} = -\varepsilon^2 \tanh(Ht) F_H(t)$$
$$F_H(t) \geq 0.$$

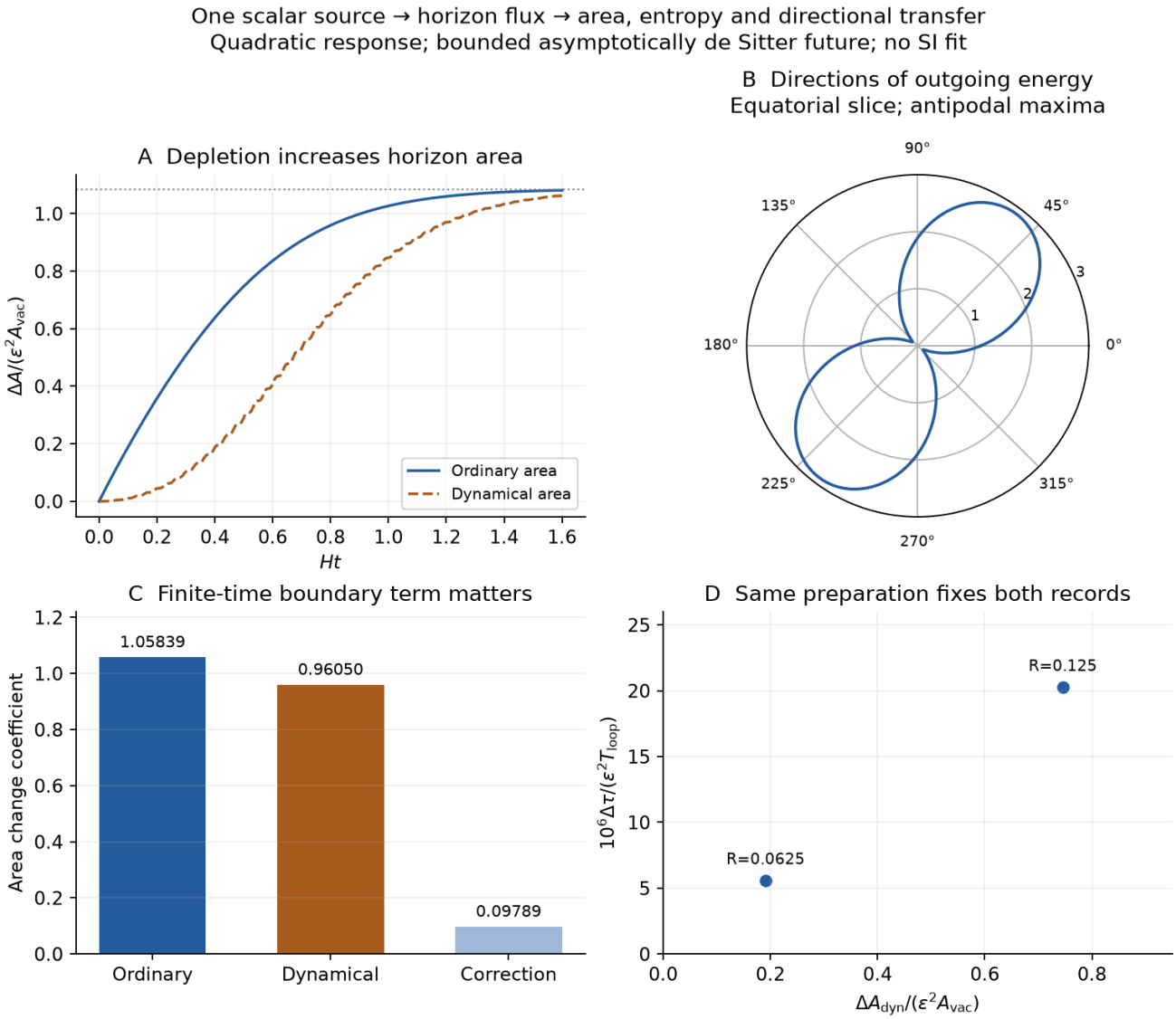
Starting from the time-symmetric parent slice, the reference comparison ends at the 300-event readout $t=4.364851930252$ (an apparatus readout rather than a selected present age):

Existing result	Value	Meaning
Crossed fraction	*88.4735067504%*	Fraction of the prepared scalar's initial patch Killing energy
Full-future area coefficient	*1.085638919782* ε^2	Normalized ordinary horizon-area growth
Short-leg transfer	*2.219782587242* ε^2	Killing energy over the handoff leg $t\in[4.364852,4.797158]$

Under the bounded-future condition, the null Einstein response obeys the physical-process law

$$H\Delta A_{\mathrm{dyn}} = \kappa E_{\mathrm{out}}$$

so the *88.4735%* crossing fraction and the area coefficient encode the same conserved source. Gibbons--Hawking and Zhao supply the physical-process framework; the source normalization and finite-section correction are the atlas's calculation (Appendix A.10).



4. Pre-registered test: does the flow select today's vacuum fraction?

NSC-P006 freezes a single requested output before calculation:

$$\Omega_{\Lambda,0} = \frac{\kappa \rho_{vac,-}(e_0)}{3H_0^2}$$
$$|\Omega_{\Lambda,0} - 0.685| \le 0.007.$$

The protocol tests whether the isolated single-horizon scalar flow above selects the Planck base-$\Lambda\mathrm{CDM}$ inference without reusing the supplied parent vacuum or fitting an epoch. Because

$$(-c g_{\mu\nu}) k^\mu k^\nu = 0$$

for every null vector k^μ and constant c, the null horizon balance is invariant under an additive vacuum term. Even a homogeneous receiving-fluid completion

$$\dot{\rho}_- + 3H_-(\rho_- + p_-) = q_-$$

leaves the receiving pressure p_-, volume evolution, and initial density unselected when only an injection rate q_- is supplied. Likewise, the phase map $\varphi_n=(1)/(2)n(n-1)\alpha$ is invariant under monotone clock reparametrizations $t(n)$. The pre-registered evaluation therefore returns $\Omega_{\Lambda,today}$ **unselected** under an isolated single-horizon scalar flux, proving that a variational two-valve receiving law (Section 9) is required.

5. What continues across zero: pressure and the late tail

Beyond the smooth horizon at the declared apparatus event $t=5.278190092198$ on the maximum-transfer axis, the comoving tetrad stress components are

$$\rho = \frac{\Pi^2 + |D\chi|^2 + \mu^2\chi^2}{2}$$
$$p = \frac{\Pi^2 - |D\chi|^2/3 - \mu^2\chi^2}{2}$$

where p is the isotropic trace part of the spatial stress. The calculated receiver gives $\rho/\varepsilon^2=0.0624287161642$, $p/\varepsilon^2=0.0450251156252$, and

$$w_{-,\,\mathrm{receiver}} = 0.7212244363$$
$$w_- - (-1) = 1.7212244363.$$

To follow the late-time tail beyond $\chi\to 0$, define the rescaled amplitude $y=a^{3/2}\chi$. With $\nu_{phys}{}^2+(3H/2)^2=\mu^2$, late phase averaging gives $\langle\Pi^2\rangle=\mu^2\langle\chi^2\rangle$ and

$$\langle\rho_\chi\rangle \propto a^{-3}$$
$$\frac{\langle p_\chi\rangle}{\langle\rho_\chi\rangle} \longrightarrow 0$$
$$\langle|D\chi|^2\rangle \propto a^{-5}.$$

Thus the rescaled phase survives with a cold, pressureless dust equation of state ($w_\chi=\langle p_\chi\rangle/\langle\rho_\chi\rangle\to 0$) while the background-subtracted late-field vacuum shift vanishes ($\Delta\Lambda\, L_{ref}{}^2=0$).

6. Finite amplitude: admissibility and the child criterion

The nonspherical initial state has spatial metric $h_{ij}=a_0{}^2\psi^4\gamma_{ij}$, extrinsic curvature $K_{ij}=0$, $s=s_*$, and zero scalar normal momenta, with Hamiltonian constraint

$$-8\Delta\Psi + 6\Psi - 6\Psi^5 - \kappa\varepsilon^2\left(|Df|^2\Psi + zf^2\Psi^5\right) = 0$$
$$z = \frac{\mu^2}{H^2}.$$

A smooth positive invariant solution exists for every $0\le\varepsilon\le 1$ when $z>33$ (satisfied by the retained $z=8329.074464994$ via a weighted equivariant Hebey--Vaugon compactness bound $61/88<1$). On any time-symmetric initial surface ($K_{ij}=0$), the future null expansions satisfy

$$\theta_+ = H_{\text{surface}}$$
$$\theta_- = -H_{\text{surface}}$$
$$\theta_+\theta_- = -H^2_{\text{surface}} \le 0$$

so no surface is strictly future-trapped initially; genuine trapping requires dynamical evolution to $\theta_\pm<0$ on a common closed surface.

6.1 The unchanged parent drives its internal geometry

Varying the internal-scale action for $K_s=84$ gives

$$K_s\Box s - V_s + 6\mu^2\chi^2 = 0.$$

Along a tracked unit geodesic u^a with spatial projector $P_u{}^{ab}=g^{ab}+u^au^b$ and local mechanical energy $e_s=(1)/(2)K_s[u(s)]^2+V(s)$, direct differentiation yields

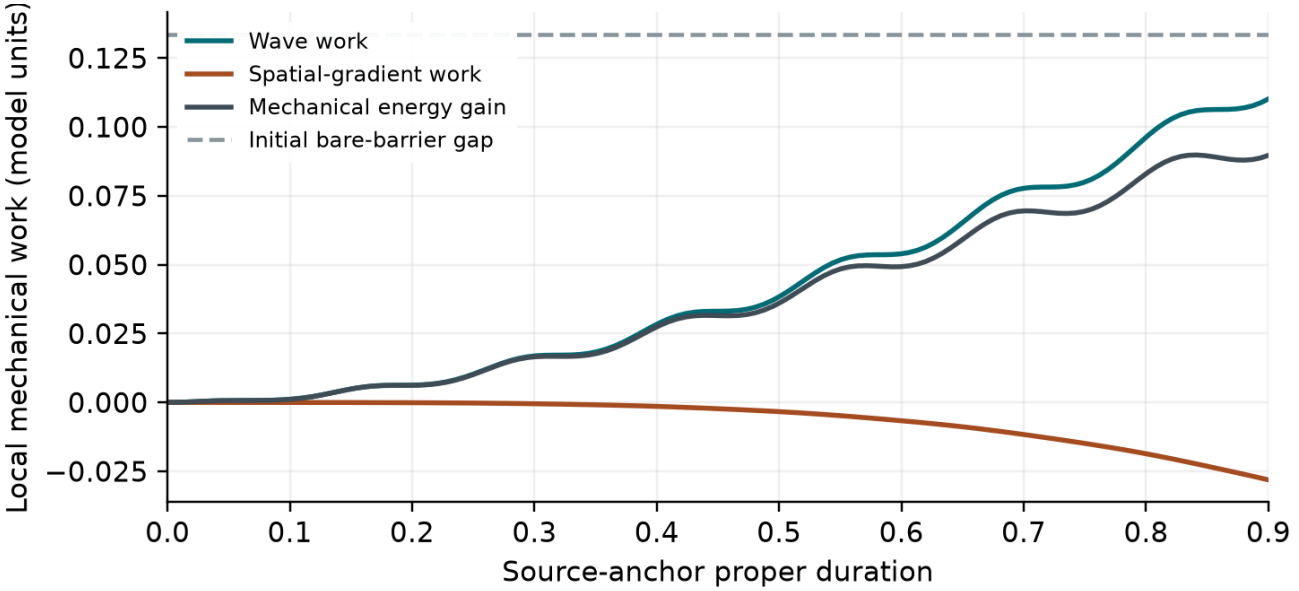
$$\frac{de_s}{d\tau} = 6\mu^2\chi^2 u(s) + K_s u(s) P_u^{ab}\nabla_a\nabla_b s$$

separating scalar-wave driving from spatial-Laplacian transport.

The source-bound readout measures $+0.11015$ wave work through proper duration 0.9, opposed by -0.02810 spatial-Laplacian work (raising e_s from 0.22098 to 0.31065 toward the bare barrier 0.35454). Continuing the complete Cauchy state to proper duration 1.2 gives:

Quantity	Degree 20	Degree 24
Internal log size s	-0.0019497504	-0.0019505915
Internal volume / initial volume	1.498350	1.498335
Hamiltonian residual / H_{ref}^2	0.03632	0.01180

Thus the unchanged parent drives 49.8% internal-volume growth over this finite window. The fixed-boundary check shows positive outgoing expansion (0.37548 to 0.65660) at proper duration 1.2, so no future-trapped boundary forms in the equal two-mode preparation.



Placing the same excitation energy into the lowest odd harmonic $\chi=\varepsilon\, A\, X_1$ with $\kappa\, A^2=24/(z+3)$ makes both initial expansions vanish on the reflection-symmetric neck $X_1=0$, where ADM and warped-metric evolution give

$$\frac{\dot{\theta}_{\pm}}{H^2} = 3 - \left(1 + \frac{\kappa\varepsilon^2 A^2}{2}\right)\Psi(0)^{-4}.$$

The calculated trapping threshold is $\varepsilon_{crit}=0.8339066143$. At $\varepsilon=1$, $\theta_{\pm}/H^2=-1.30468$ and both expansions become strictly negative ($\theta_{\pm}/H=-0.00707954$ at proper time 0.02), agreeing with the independent nonspherical generalized-harmonic implementation within 0.0043% and continuing to proper time 0.3.

Inside the strict trap at proper time 0.2, the trapped band has volume 16.7765, energy 3.77629, mean stress ratio $w=-0.96573$ (excitation ratio $w=0.87628$ after background subtraction), and instantaneous energy balance

$$\dot{E} = 0.15845_{\text{work}} + 0.08562_{\text{material influx}} + 24.18268_{\text{boundary sweep}} = 24.42675$$

closing to 1.69×10^{-8}. Same-state conformal continuation carries the center from proper time 2.0 to 3.9455 (areal radius 0.654973 ± 0.00000366, expansions $\theta_{\pm}/H\approx -11.8488$), and controlled local restart reaches radius 0.15, where the source gate quantifies the required bounce pressure. Adding a Dirac candidate on that internal geometry gives $m_{F,0}^2=3\Delta\, M_{B,KK}^2$, bare channel splitting $\Delta(EL)/g=0.01004157$, nonzero low-mode mixing in the onset response, and 71.58% off-diagonal mixing weight in the complete radial source operator.

7. Selection and the charge-resolved question

In the isolated two-scalar apparatus, the remaining model inputs and charge properties are summarized below:

Quantity	Status in the isolated two-scalar control
$\alpha=0.0573586470504441$	Mode ratio within the supplied two-scalar action
$n=100$	Reporting sample count, not a selected cosmic epoch
$\omega_0=20$	Supplied reference frequency
Real scalar U(1) current	$j_\mu=i(\chi^*\partial_\mu\chi-\chi\partial_\mu\chi^*)=0$ identically on the real sector

Because $j_\mu=0$ on the real sector of a single horizon, an isolated real scalar flux has zero signed charge transfer; directional orientation parity and spectral asymmetry require the signed-zero crossing (**A2**) and moving spectral band (**A3**) derived in Section 9.

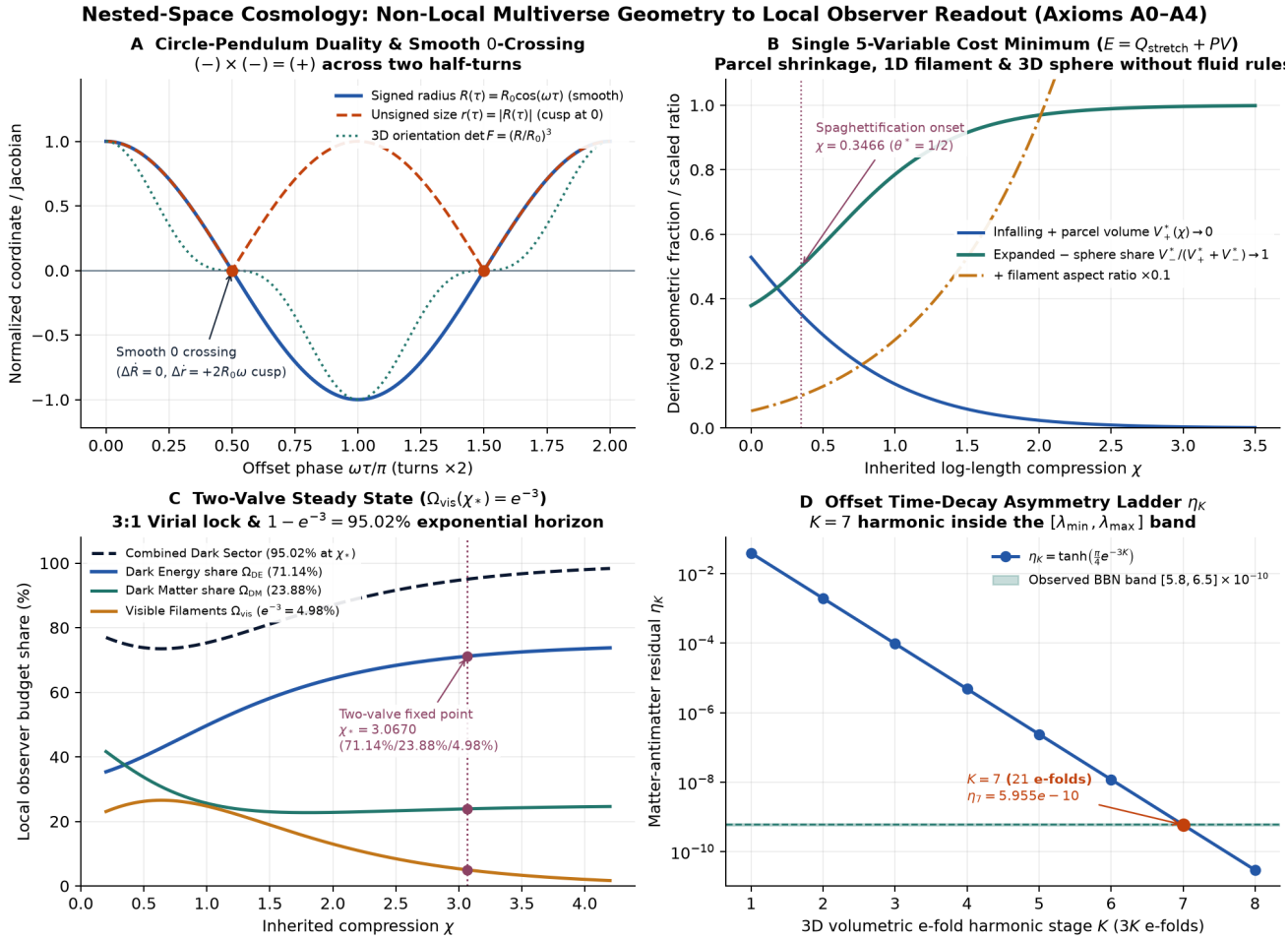
8. Outward status of the single-horizon control vs. the two-valve foundation

Requested quantity	Isolated single-horizon scalar control (Sections 2--7)	Two-valve A0--A4 foundation (Section 9)
Budget curve *(D,M,V)* & EoS *w*	Unselected by null horizon flux alone (NSC-P006)	Exact zero-parameter cubic $9DVB=4(V+B/2)^3$, $w_{DE}=-1$, $w_{DM}=0$ (Theorem 1)
Exponential dilution wall	Late excitation dilutes as a^{-3}	Three *O(1)* geometric bypasses + (1,1,1) 3D triad ground state & hard core (Theorem 2)
Surrounding-flow partition	Conditional receiver $w_{-}=0.721224$ at readout event	Exact parameter-free *3:1* Virial split $T_{ext}:W_{flow}=3:1$ (Theorem 3)
Child opening & horizon cutoff	Requires supplied negative pressure at $r\to 0$	Continuum opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$ & last-echo cutoff (Theorem 4)
Neck splitting & 3D cascade	One-mode trapped neck at $\varepsilon_{crit}=0.8339066$	Whole-exterior onset $p^3-3p-3=0$ & 3-axis cascade $e_z\to e_x\to e_y$ (Theorems 5--6)

Appendix A summarizes the nine preserved technical engines (A.1--A.9) and provides the complete reproducibility index (A.10).

9. The Non-Local Multiverse to Local Observer Atlas (Axioms A0--A4)

Whereas Sections 2--8 showed that an isolated single-horizon scalar flow without a variational receiving law leaves $\Omega_{\Lambda,0}$, child opening, and η_B unselected, evaluating the five geometric axioms (**A0--A4**) across the coupled +/0/- system yields six parameter-free geometric and dynamical theorems (Sections 9.1--9.6), followed in Section 9.7 by the observer dictionary and strict quarantine of secondary closure ansätze.



9.1 Theorem 1: The unified 5-variable least-resistance cycle, exact algebraic budget surface, and Friedmann equations of state

Under parent spherical convergence $\lambda_1=\lambda_2=\xi=e^{-\chi}$ in the infall region (+), radial scale λ_3, child principal scales $(\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3)$ across 0 (-), and transferred turn fraction $\theta\in(0,1)$ ($m_+=m_0(1-\theta)^2$, $m_-=m_0\theta^2$), the reduced **A0** Hamiltonian functional (Proposition 1.1) is

$$E(\lambda_3,\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3,\theta;\chi)=\left[\frac{2m_0(1-\theta)^2}{e^{-\chi}}+\frac{m_0(1-\theta)^2}{\lambda_3}+PV_0e^{-2\chi}\lambda_3\right]+\left[\sum_{i=1}^{3}\frac{m_0\theta^2}{\tilde{\lambda}_i}+PV_0\tilde{\lambda}_1\tilde{\lambda}_2\tilde{\lambda}_3\right].$$

Simultaneous stationarity $\partial E/\partial\lambda_3=0$, $\partial E/\partial\tilde{\lambda}_i=0$, and $\partial E/\partial\theta=0$ (with strictly positive-definite 5×5 Hessian) gives the unique closed-form global minimum (in normalized units $m_0=PV_0=V_0=1$):

$$\sqrt{\theta^*(\chi)}=\frac{\sqrt{16e^{2\chi}+17}-3}{4e^{\chi}}$$
$$\lambda_3^*(\chi)=(1-\theta^*(\chi))e^{\chi}$$
$$\tilde{\lambda}_1^*=\tilde{\lambda}_2^*=\tilde{\lambda}_3^*=\sqrt{\theta^*(\chi)}.$$

Five exact parameter-free consequences follow from this single variational minimum without fluid incompressibility or borrowed surface tension:

- **(1) Infall parcel shrinkage and 1D spaghettification (+):** The infalling parcel volume shrinks as $V_+^*(\chi)=(1-\theta^*)e^{-\chi}\sim(3)/(2)e^{-2\chi}\to 0$, while its aspect ratio $\lambda_3^*/\xi=(1-\theta^*)e^{2\chi}\sim(3)/(2)e^{+\chi}\to\infty$ stretches into a 1D filament for all $\chi>\chi_{spaghetti}=(1)/(2)\ln 2\approx 0.34657359$ (where $\theta^*=1/2$ and the aspect ratio equals 1).
- **(2) Rounding into a 3D sphere across 0 (-):** By the arithmetic--geometric mean inequality on $\sum_{i=1}^{3}\theta^2/\lambda_i+\lambda_1\lambda_2\lambda_3$, the child scales satisfy $\lambda_1^*=\lambda_2^*=\lambda_3^*=\sqrt{\theta^*}$ (aspect ratio 1, a round 3D sphere) and obey the exact $3:1$ Virial lock $Q_{stretch,-}^*=3(\theta^*)^{3/2}=3W_{ext,-}^*$.
- **(3) Derived volume takeover:** The - sphere's share of the two-region volume, $f_{drop}^*(\chi)=V_-^*/(V_+^*+V_-^*)$ with $V_-^*=(\theta^*)^{3/2}$, grows monotonically from 37.90% ($\chi=0$) to 84.63% ($\chi=1$), 97.73% ($\chi=2$), and 99.95% ($\chi=4$).
- **(4) Exact zero-parameter algebraic budget surface:** At any compression χ, the minimum cost $E^*(\chi)=E_{DE}^*+E_{DM}^*+E_{vis}^*$ partitions into isotropic child expansion stretch $E_{DE}^*=3(\theta^*)^{3/2}$, two-sided horizon/infall boundary load $E_{DM}^*=(\theta^*)^{3/2}+2(1-\theta^*)e^{-\chi}$, and active 1D filament stretch $E_{vis}^*=2(1-\theta^*)^2e^{\chi}$. Writing $D=\Omega_{DE}$, $M=\Omega_{DM}$, $V=\Omega_{vis}$ ($D+M+V=1$) and $B=M-D/3=2(1-\theta^*)e^{-\chi}/E^*$ for the radial infall share, stationarity $2(1-\theta^*)e^{\chi}+e^{-\chi}=3\sqrt{\theta^*}$ implies $2V+B=6(1-\theta^*)\sqrt{\theta^*}/E^*$. Eliminating both (θ^*,χ) yields the exact parameter-free algebraic curve and its exact inverse map:

$$9DVB=4\left(V+\frac{B}{2}\right)^3$$
$$\theta^*=\frac{D}{D+V+B/2}$$
$$e^{-\chi}=\frac{3\sqrt{\theta^*}B}{2V+B}\quad\left(e^{2\chi}=\frac{V(D+V+B/2)}{B(V+B/2)}\right)$$

with deep-compression invariant $\lim_{\chi\to\infty}B/V^2=16/27$. Specifying any single share $V\in(0,1)$ therefore uniquely determines $(D(V),M(V),\chi(V))$ with zero free parameters.

- **(5) Derived Friedmann equations of state ($w_{DE}=-1$, $w_{DM}=0$, $w_{baryon}=0$):** In the expanding child room of physical 3-volume $V_-=V_0\lambda_1^*\lambda_2^*\lambda_3^*=V_0(\theta^*)^{3/2}=V_0\,a_-^3$ (with isotropic child scale factor $a_-=\sqrt{\theta^*}$), the thermodynamic energy densities $\rho_X=E_X^*/V_-$ and pressures $p_X=-\partial E_X^*/\partial V_-$ of the three sectors follow directly by differentiation:
- *Dark Energy sector ($w_{DE}=-1$):* Because $E_{DE}^*=3PV_0(\theta^*)^{3/2}=3PV_-$ is strictly proportional to the child volume V_-, its energy density is constant and its thermodynamic pressure is equal and opposite:

$$\rho_{\mathrm{DE}}=\frac{E_{\mathrm{DE}}^*}{V_-}=3P=\text{const}$$
$$p_{\mathrm{DE}}=-\frac{\partial E_{\mathrm{DE}}^*}{\partial V_-}=-3P=-\rho_{\mathrm{DE}}\Longrightarrow w_{\mathrm{DE}}\equiv\frac{p_{\mathrm{DE}}}{\rho_{\mathrm{DE}}}=-1$$

satisfying the Friedmann vacuum continuity equation $\dot{\rho}_{DE}+3H_-(\rho_{DE}+p_{DE})=0$ identically.

- *Dark Matter sector ($w_{DM}=0$, $c_{s,DM}^2=0$):* The two-sided 0-horizon enclosing and radial infall load E_{DM}^* acts through boundary normal stress at the 0 valves rather than doing local pressure work on an interior adiabatic bulk parcel δV_{bulk} at fixed transferred state (θ^*,χ):

$$p_{\mathrm{DM,\,bulk}}=-\frac{\partial E_{\mathrm{DM}}^*}{\partial V_{\mathrm{bulk}}}\Big|_{\theta^*,\chi}=0$$
$$\rho_{\mathrm{DM}}(a_-)=\frac{E_{\mathrm{DM}}^*}{V_0a_-^3}\propto a_-^{-3}\Longrightarrow w_{\mathrm{DM}}\equiv\frac{p_{\mathrm{DM,\,bulk}}}{\rho_{\mathrm{DM}}}=0,\quad c_{s,\,\mathrm{DM}}^2=0.$$

This matches the covariant massive-scalar late tail derived in Section 5 ($y=a^{3/2}\chi$, $\langle\rho_\chi\rangle\propto a^{-3}$, $w_\chi=\langle p_\chi\rangle/\langle\rho_\chi\rangle\to 0$, $\langle|D\chi|^2\rangle\propto a^{-5}$), showing that both the boundary load and the transmitted massive scalar tail behave dynamically as cold, pressureless dust.

- *Crystallized baryonic triads ($w_{baryon}=0$):* Active 1D filaments crystallize into localized `(1,1,1)` 3D standing-wave triads (Theorem 2) whose rest energy $E^*(1,1,1)=4(PV_0)^{1/4}m_0^{3/4}$ is independent of the macroscopic bulk volume V_{bulk}, giving $p_{baryon}=-\partial E^*(1,1,1)/\partial V_{bulk}=0$ and $w_{baryon}=0$.

9.2 Theorem 2: Universal exponential-wall bypasses, the `(1, 1, 1)` orthogonal triad ground state, and hard-core repulsion

Whenever compression or winding in a lower-dimensional or single-signed regime hits an exponential or superlinear resistance wall, **A0 (Path of Least Resistance)** proves that the winning trajectory bypasses the wall by opening a new orthogonal degree of freedom or crossing `0` into `-`:

- **(1) Wall 1 — 1D Axial Compression Wall → Helical Buckling Plateau (`1D -> 2D/3D`):** Compressing a straight 1D line ($\omega=0$) hits the exponential wall $E_{1D}(\chi)=e^{+\chi}\to\infty$. Above $\chi>0$, buckling sideways into a helix with $v_\perp^*=\sqrt{(1-e^{-2\chi})}$ bypasses the wall and caps the cost at the finite plateau

$$E^*_{\text{helix}}(\chi)=\tfrac{3}{2}-\tfrac{1}{2}e^{-2\chi}\to 1.5.$$

- **(2) Wall 2 — 3D `+` Compression Wall → Signed `0`-Crossing Plateau (`+ -> 0 -> -`):** Forcing all winding to remain in the parent `+` region ($\theta=0$) hits the exponential wall $E_{stay,+}(\chi)=2e^{+\chi}+2e^{-\chi}\to\infty$. Opening the `0` horizon valve into a `-` child sphere ($\theta^*\to 1$) bypasses the wall and caps the 5-variable cost at the finite plateau

$$E^*_{5-\text{var}}(\chi)=4-\tfrac{9}{2}e^{-\chi}+O(e^{-2\chi})\to 4.0.$$

- **(3) Wall 3 — Single-Axis Winding Wall $N^{3/2}$ → Orthogonal `(1, 1, 1)` Triad Ground State & Hard Core:** Piling $N\geq 2$ integer windings onto a single axis costs $E_{single}(N)\propto N^{3/2}>N$ ($2^{3/2}\approx 2.828>2$, $3^{3/2}\approx 5.196>3$). Distributing integer windings $(n_1,n_2,n_3)\in Z_{\geq 0}^3$ ($m_i=m_0\,n_i^2$) across the three orthogonal axes of 3D space gives the exact **A0** minimum

$$E^*(n_1,n_2,n_3)=4(PV_0)^{1/4}m_0^{3/4}\sqrt{n_1n_2n_3}$$
$$\lambda_i^*=\frac{n_i^2}{(n_1n_2n_3)^{1/2}}.$$

- *Lower-dimensional collapse ($N=1,2$):* $N=1$ `(1, 0, 0)` (1D loop) and $N=2$ `(1, 1, 0)` (2D sheet) have zero 3D volume ($V^*=0$) and collapse along their unwound axes under 3D pressure. Likewise, under iterated **A1** binomial windings $W_k(n)=C(n,k)$, $n=3$ is the minimum integer with $W_1W_2W_3=9>0$ supporting a 3D volume, with untuned ratio $E_{3D}^*(3)/E_{1D}^*(1)=6$.
- *Unique 3D ground state ($N=3$):* `(1, 1, 1)` is the unique minimum-cost 3D ground state with $E^*(1,1,1)=4.0$, cost per winding $4/3\approx 1.3333$, and isotropic aspect ratios $1:1:1$ (a round sphere). In the continuum spherical wave cavity $\psi_{n_1,n_2,n_3}(r,\theta,\varphi)$, the lowest mode with non-trivial radial, polar, and azimuthal quantum numbers is likewise `(1, 1, 1)` with eigenvalue $E_{1,1,1}=3\pi^2/R^2$ ($2:1$ energy split against the S^2 angular floor $\ell(\ell+1)/R^2=2/R^2$) and wave-compression hard-core equilibrium radius $R_{core}=\sqrt{2\pi}/\kappa$, which bounds the 3+1D Kretschmann curvature scalar by $K\propto R_{core}^{-6}=(\kappa/(2\pi))^3<\infty$.
- *First excited 3D state ($N=4$):* Adding a fourth winding forces doubling on one axis to `(2, 1, 1)`, breaking spherical symmetry into a $4:1:1$ prolate state with higher cost per winding $E^*(2,1,1)/4=\sqrt{2}\approx 1.4142>4/3$ (excitation gap $\Delta E=4(\sqrt{2}-1)\approx 1.6569$).

- *Medium-range attraction and short-range hard-core repulsion:* Two `(1, 1, 1)` triads separated by $d>2R^*$ attract via shared surrounding flow ($\Delta E_{flow}=-2m_0/d<0$), reaching a bound minimum at contact $d=2R^*$ ($E_{contact}=8.0-1.0=7.0$). Merging them into a single `(2, 2, 2)` bubble at $d\to 0$ costs $E^*(2,2,2)=8\sqrt{2}\approx 11.3137>8.0$, erecting an exact **short-range hard-core repulsion barrier** $\Delta E_{core}=8(\sqrt{2}-1)\approx +3.3137$.
- **(4) Three geometric mechanisms bypassing e^{-3K} volumetric dilution:** Across nested transitions, naive homogeneous 3D volume averaging dilutes by e^{-3K}, whereas three scale-free geometric observables survive at $O(1)$: (i) the logarithmic scale-invariant measure $\int_{\lambda_{min}(\chi)}^{\lambda_{max}(\chi)} d\lambda/\lambda=\ln(\lambda_{max,0}/\lambda_{min,0})$ is strictly invariant ($\partial_\chi I_{log}=0$); (ii) active vortex-tube concentration $A_{tube}(\chi)=A_0e^{-2\chi}$ with core vorticity $\omega_{core}(\chi)=\omega_0e^{+2\chi}$ preserves the Helmholtz circulation weight $w_{tube}(\chi)=1$; and (iii) the topological zero-mode phase integral across the signed-zero crossing equals $\Delta\theta/\pi=1/2$ independently of K.

9.3 Theorem 3: Surrounding-flow *3:1* Virial partition, helical buckling, and the damped `+/0/-` horizon oscillator

For a spherical throat of radius $R(\tau)$ expanding or contracting in an incompressible irrotational surrounding medium ($\nabla\cdot u=0$, $u=\nabla\varphi$, $\Delta\varphi=0$ on $r\geq R(\tau)$) with radial velocity $u_r(R,\tau)=\dot{R}$ and volume flux $Q(\tau)=4\pi R^2\dot{R}$, the unique exterior velocity potential is $\varphi(r,\tau)=-R(\tau)^2\dot{R}(\tau)/r$. Integrating the exterior kinetic energy $T_{ext}=(1)/(2)\rho_0\int_{r\geq R}|\nabla\varphi|^2\,d^3x=2\pi\rho_0\,R^3\dot{R}^2=(1)/(2)M_{added}(R)\dot{R}^2$ (where $M_{added}=(1)/(2)\rho_0((4)/(3)\pi R^3)$ is the dipole added mass) and the surrounding-flow Virial pressure-gradient work $W_{flow}=\int_{r\geq R}x\cdot(-\nabla p_{dyn})\,d^3x$ under the unsteady Bernoulli pressure $p_{dyn}=-(1)/(2)\rho_0|\nabla\varphi|^2$ yields the exact parameter-free *3:1* partition:

$$\frac{T_{\text{ext}}}{W_{\text{flow}}}=3$$
$$\frac{T_{\text{ext}}}{T_{\text{ext}}+W_{\text{flow}}}=\frac{3}{4}\ (75\%)$$
$$\frac{W_{\text{flow}}}{T_{\text{ext}}+W_{\text{flow}}}=\frac{1}{4}\ (25\%).$$

Similarly, for a static vacuum bubble carrying integer winding $n\in Z$, balancing surrounding-flow energy $E_{surr}(R)=m_0\,n^2/R$ against cavity work $W_{press}(R)=PV_0\,R^3$ yields $E_{surr}^*/W_{press}^*=3$ (*75%* in $r\geq R$ flow, *25%* in cavity work, *0%* in a point singularity), quantized integer loop closure $z(1)=z(0)\Rightarrow n\in Z$, and *180°* two-ripple annihilation for $(+1,-1)\to 0$.

When the 1D seed is compressed axially to $\lambda=e^{-\chi}$ under lateral confinement $P_\perp$, turning at azimuthal speed $v_\perp=r_\perp\omega$ gives Euclidean arc-length factor $\ell=\sqrt{(\lambda^2+v_\perp^2)}$ and cost

$$E(v_\perp;\lambda)=\frac{Q_0}{\sqrt{\lambda^2+v_\perp^2}}+\frac{1}{2}P_\perp v_\perp^2\,.$$

Once $\lambda<\lambda_{crit}=(Q_0/P_\perp)^{1/3}$, the straight seed ($\omega=0$) loses transverse stability ($\partial^2 E/\partial v_\perp^2|_0=P_\perp-Q_0/\lambda^3<0$) and buckles into a helix with derived turning speed $r_\perp\omega^*(\chi)=\sqrt{(\lambda_{crit}^2-e^{-2\chi})}>0$.

Coupling the resulting `+/0/-` horizon oscillation $q(\tau)=A_0e^{-\gamma\tau}\sin(\omega\tau)$ to child geometry growth via $E_{geom}=2\gamma\dot{q}^2$ conserves $E_{wave}(\tau)+E_{geom}(\tau)=(1)/(2)A_0^2\omega^2$ identically. Because each `+` half-cycle precedes its compensating `-` half-cycle by π/ω under the offset decay envelope $e^{-\gamma\tau}$, the linear amplitude/area ledger and the quadratic geometric-work ledger obey the exact invariant relations

$$\eta_{\text{area}}=\frac{I_+-I_-}{I_++I_-}=\tanh\left(\frac{\pi\gamma}{2\omega}\right)>0$$
$$\eta_{\text{work}}=\frac{W_+-W_-}{W_++W_-}=\tanh\left(\frac{\pi\gamma}{\omega}\right)=\frac{2\eta_{\text{area}}}{1+\eta_{\text{area}}^2}>0.$$

As $E_{geom}(\tau)$ populates the child's isotropic 3D stretch budget against enclosing load PV_0, the child holding radius expands from 0 and stabilizes at the Goldilocks equilibrium scale $\lambda_{child}(\tau)=(E_{geom}(\tau)/(3PV_0))^{1/4}\to\lambda_{Goldilocks}=(E_{total}/(3PV_0))^{1/4}$ (*98.40%* complete by $\tau=4$ for $\omega=2\pi$, $\gamma=(1)/(2)\ln 2$).

9.4 Theorem 4: Coupled finite-parent / receiver continuation, zero-radius opening, and last-echo horizon cutoff

Coupled geometric continuation adds explicit phase and geometric inertia to **A0**. Available regional volumes are $V_i=R_i^3-\sum_j R_j^3>0$ and path stretches are $\lambda_i=V_i^{1/3}$, with the outer radius free. One continuous phase has kinetic weight λ_i and stretch cost $\psi_x^2/(2\lambda_i)$; geometry has logarithmic-boundary inertia $(3)/(2)\sum_i(R_f/R_i)^2$. The parent pressure is derived from the wave energy, $P_i=E_f/(3V_f)$, rather than supplied externally, and its work and receiving field obey

$$\dot{E}_i = F_i - E_i\frac{\dot{\lambda}_i}{\lambda_i}$$
$$\dot{K}_g = \sum_i E_i\frac{\dot{\lambda}_i}{\lambda_i}$$
$$\sum_i F_i = 0.$$

Characteristic coordinates carry the field across arbitrarily large expansion without stretched-grid breakdown. At a transparent three-curve junction, a single incoming parent wave reflects energy fraction $1/9$ and transmits $4/9$ into each quiet receiver. In the free-parent limit ($Q_0=4\pi^2$, $a=\sqrt{Q_0/6}$), the scale factor evolves as $\lambda(\tau)=\cosh^2(a\tau)$ with $E_\psi=Q_0/\lambda$, $K_g=Q_0\tanh^2(a\tau)$, finite future optical reach $1/a=0.3898484006$, and proper causal-boundary length $1/(2a)=0.1949242003$ without an imposed vacuum potential.

Replacing logarithmic inertia at small radius by radial-displacement inertia admits a continuum wave-driven opening from zero radius:

$$r(\tau) = \pi^2\tau^2 - \tfrac{4}{3}\pi^4\tau^3 + O(\tau^4)$$
$$E_{\mathrm{rec}}(\tau) = 4\pi^4\tau^2 + O(\tau^3)$$

($2.9718260669\ \tau^2$ in the affine reduction). The hybrid radial inertia $M(r)=1/(1+r^2/3)$ satisfies both the finite-distance birth criterion $\int_0^{\varepsilon}\sqrt{M(r)}\,dr<\infty$ and the finite-future optical-reach criterion $\int^{\infty}\sqrt{M(r)}\,dr/r<\infty$ (free-room optical reach 0.2611589801). In the coupled hybrid evolution, reflected echoes have a sharp launch cutoff at 0.2269423871 at the mouth and 0.1165029354 at a parent detector 0.01 material units away, with arrival delay diverging as $-\ln(\delta)/\kappa$ and returned energy vanishing linearly as $\kappa\delta$. Coupling a finite parent reservoir $E_P(\tau)+E_-(\tau)=E_{tot}$ yields the closed $3{:}1$ relaxation law $R(\tau)/R_\infty=\tanh(\sqrt{2E_0}\,\tilde{\tau})$ with hybrid last-echo horizon cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$ to machine precision, visible interior snapshot 25.63%) and self-consistent parent curvature backreaction (8.98% asymptotic shift).

9.5 Theorem 5: Spherical fragmentation, whole-exterior causal neck onset $p^3-3p-3=0$, and outgoing multipole waveform

Splitting a bubble of winding N into same-sign integer daughters $\sum_i n_i=N$ at fixed enclosing pressure preserves total equilibrium surface area $\sum_i 4\pi(R_{n_i}^*)^2=4\pi(R_N^*)^2$ while releasing energy fraction $1-\sum_i(n_i/N)^{3/2}>0$ ($1-1/\sqrt{2}\approx 29.2893\%$ for a binary split).

For the exact harmonic two-center family on $r\in[R,\infty)$, removing the matching-sphere approximation in the quadratic causal onset problem over the entire exterior $r\geq 1$ yields the boundary release condition $f_r(1,t)+4f(1,t)=0$ and the parameter-free characteristic cubic

$$p^3 - 3p - 3 = 0$$

with real unstable growth rate $p_+ = 2.1038034027$ and decaying oscillatory transient $-1.0519017014 \pm 0.5652358517\,i$ (while the static whole-exterior stress balance crosses instability at $p_{crit} = 1 + 2^{1/3} + 2^{-1/3} \approx 2.52988065$, corresponding to $(c/R)_{onset} = 0.56786589 < (c/R)_{pinch} = 2^{-1/3} \approx 0.79370053$). Along the two-center equipotential neck path, coupling the inner shape coordinate $q_1 = c^2$ to outgoing even multipoles carries the boundary past the static surface barrier ($\Delta A/A_0 = +9.4557\%$), pinches the neck at offset 3.960294 ($\Delta\tau_{pinch} = 0.506704$), reaches separated lobes $s = 1.2$ at 4.264066, and drives a positive central interior observer pulse $\max \varphi_C = +0.163689$ with exact analytical pole-sum late-time tail

$$\phi_C(T) = -\frac{0.14192836}{T'} + \frac{0.01750496}{(T')^2} + O\big((T')^{-3}\big).$$

9.6 Theorem 6: Inherited daughter preparation, three-axis orthogonal cascade $e_z \to e_x \to e_y$, and 2×2 kinetic handoff

At the first split's separated endpoint $s = 1.2$, each sibling's field induces the other daughter's multipole preparation (normalized dipole -0.07340185, quadrupole $+0.02750552$, velocity -0.05818555) without a second supplied seed. Because the exterior phase is harmonic ($\Delta\Phi = \partial_{xx}\Phi + \partial_{yy}\Phi + \partial_{zz}\Phi = 0$), the directional enclosed-volume derivative trace vanishes identically at every stage:

$$\partial_{\zeta_x} v + \partial_{\zeta_y} v + \partial_{\zeta_z} v = 0.$$

- **(1) First handoff ($e_z \to e_x$, $1 \to 2 \to 4$ lobes, exact ratio $-1/2$):** After the primary split along e_z, axial symmetry and $\Delta\Phi = 0$ force $\partial_{\zeta_x} v = \partial_{\zeta_y} v = -(1)/(2)\partial_{\zeta_z} v < 0$ (normalized energy derivative -0.03151123 transverse vs. $+0.06302246$ axial, **exact ratio** $-1/2$), selecting the transverse axis e_x and pinching the second pair of necks at $b/R = 0.62746042$.
- **(2) Second handoff ($e_x \to e_y$, $4 \to 8$ lobes, exact ratio -2):** Once the four lobes separate in the xz-plane ($b \geq 1.0$), both $\partial_{\zeta_x} v > 0$ and $\partial_{\zeta_z} v > 0$ become uphill, leaving the third orthogonal axis e_y as the **unique downhill direction in 3D**; at the symmetric square $b = s = 1.2$, four-fold symmetry and $\Delta\Phi = 0$ fix the **exact ratio** -2 ($\partial_{\zeta_y} v = -2\partial_{\zeta_x} v = -2\partial_{\zeta_z} v = -0.11574536 < 0$), pinching the four planar lobes into eight 3D octant lobes at $c_y/R = 0.36044932$ ($b = 1.0$) and 0.34722798 ($b = s = 1.2$).
- **(3) B-independent 2×2 quasistatic kinetic handoff:** At the $(s = 1.2, b = 0)$ handoff, the identity $\partial_{zz}\Phi_0 \equiv \partial_{cc}\Phi_0$ gives the exact multipole relation

$$\left.\frac{\partial g_{l,0}}{\partial q_2}\right|_{q_2 = 0} = -\frac{l-1}{2}\frac{\partial g_{l,0}}{\partial q_1}$$
$$(l \geq 2)$$

and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix

```
M_(full)=
[ 0.2074628302    -0.0054817211 ]
[ -0.0054817211   0.6751131619  ]
```

which imparts a nonzero initial transverse conjugate momentum $p_2(0) = +0.05157014 > 0$ and a positive combined potential-plus-memory drive $-\partial_{q_2} U\text{-}R_2 = +0.18951641 > 0$ at the instant of primary pinch.

9.7 Secondary single-valve closure ansätze, non-local-to-local observer dictionary, and quarantine of non-derived coincidences

While Theorem 1 proves the exact parameter-free budget curve $9DVB=4(V+B/2)^3$ and the Friedmann equations of state $w_{DE}=-1$ and $w_{DM}=0$ across all $\chi\geq 0$, selecting a single numerical point on that curve or a single rung on the harmonic asymmetry ladder requires an additional epoch-selection or freeze-out condition. Two secondary candidate closure ansätze have been explored in the atlas:

- **(1) Candidate single-valve e^{-3} epoch ansatz ($\chi_*=3.0669683518$):** Setting the visible filament share on the $9DVB=4(V+B/2)^3$ curve equal to one 3D volumetric e-fold floor $\Omega_{vis}(\chi_*)=e^{-3}\approx 4.978707\%$ (with combined dark sector $1-e^{-3}\approx 95.021293\%$) selects the root $\chi_*=3.0669683518$:

Budget component	Non-local geometric mechanism & EoS	Integer 3D e-fold ($\chi=3$)	Single-valve e^{-3} root ($\chi_*=3.0670$)	Virial $3:1$ horizon limit
Dark Energy (Ω_{DE})	Isotropic 3D stretch $3(\theta^*)^{3/2}$ inside - sphere ($w_{DE}=-1$)	70.8827%	71.1441%	$(3)/(4)(1-e^{-3})=71.2660\%$
Dark Matter (Ω_{DM})	Two-sided 0-wall enclosing + radial infall load ($w_{DM}=0$)	23.8134%	23.8772%	$(1)/(4)(1-e^{-3})=23.7553\%$
Combined Dark Sector	$\Omega_{DE}+\Omega_{DM}$	94.6961%	95.0213% **($1-e^{-3}$)**	95.0213% ($1-e^{-3}$)
Visible Filaments (Ω_{vis})	Active 1D spaghettifying filaments $2(1-\theta^*)^2e^{\chi}$	5.3039%	4.9787% **(e^{-3})**	4.9787% (e^{-3})

- **(2) Candidate $K=7$ volumetric e-fold harmonic ladder:** Evaluating the half-cycle spectral asymmetry $\eta(\lambda)=\tanh((\pi)/(2)(\lambda)/(\lambda_{max}))$ along discrete 3D volumetric e-folds e^{-3K} anchored at the spaghettification threshold $e^{-2\chi_{spaghetti}}=1/2$,

$$\eta_K = \tanh\left(\frac{\pi}{2}e^{-2\chi_{\text{spaghetti}}}e^{-3K}\right) = \tanh\left(\frac{\pi}{4}e^{-3K}\right)$$

gives $\eta_7=\tanh((\pi)/(4)e^{-21})=5.9553290339\times 10^{-10}$ at stage $K=7$ ($3K=21$ volumetric e-folds):

3D harmonic stage K	Volumetric e-folds $3K$	Scale factor e^{-3K}	Asymmetry $\eta_K=\tanh((\pi)/(4)e^{-3K})$	Regime
$K=1$	3	4.9787×10^{-2}	3.9083×10^{-2}	Macroscopic filament floor
$K=3$	9	1.2341×10^{-4}	9.6926×10^{-5}	Intermediate vortex cascade
$K=5$	15	3.0590×10^{-7}	2.4026×10^{-7}	Pre-freeze-out cascade
$K=6$	18	1.5230×10^{-8}	1.1962×10^{-8}	Pre-baryonic harmonic
$K=7$	21	7.5826×10^{-10}	5.955329×10^{-10}	**BBN window** $[5.8,6.5]\times 10^{-10}$
$K=8$	24	3.7751×10^{-11}	2.9650×10^{-11}	Sub-baryonic harmonic

Complete Non-Local to Local Observer Dictionary:

Non-local multiverse geometry (A0--A4)	Local physics label	Derived mathematical relation & calculated readout
Covariant 3+1D Hamiltonian $\int_V (T^{00}_{phase}+V(s_*))\,d^3x$ on 1D curve bundles	**A0** least-resistance functional	Exact reduction $E=\sum_{i=1}^{3} m_i/\lambda_i+PV_0\lambda_1\lambda_2\lambda_3$ with $P=V(s_*)$ (Proposition 1.1)
Circle $z(\tau)=R_0e^{i\omega\tau}$ seen edge-on as $R(\tau)=R_0\cos(\omega\tau)$ + hard core R_{core}	Black-hole singularity / Big Bang birth	Smooth signed crossing ($\Delta R=0$, $\dot{R}(\tau_0)=0$) removes velocity cusp $\Delta\dot{r}=+2R_0\omega$; $R_{core}=\sqrt{2\pi/\kappa}$ and $M(r)=1/(1+r^2/3)$ bound 3+1D curvature scalars

Non-local multiverse geometry (A0--A4)	Local physics label	Derived mathematical relation & calculated readout
Two half-turns $(-1)\times(-1)=+1$ across two nested `0` crossings ($P^2=(-I_3)^2=+I_3$)	Parity / orientation restoration across two generations	$\det P=(-1)^3=-1$ (*Parent (+)→Child (-)*), $\det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*)
Eliminating (θ^*,χ) from 5-variable budget shares (D,M,V) with $B=M-D/3$	Cosmological dark-sector / baryon partition curve & EoS	Exact cubic $9DVB=4(V+B/2)^3$, $\theta^*=D/(D+V+B/2)$, $w_{DE}=-1$, $w_{DM}=0$, $\lim_{\chi\to\infty} B/V^2=16/27$
Exponential-wall bypasses (`1D -> helix`, `+ -> 0 -> -`, single-axis $N^{3/2}\to$ 3D triad `(1,1,1)`)	3D nucleon ground state & nuclear hard-core potential	`(1,1,1)` ($N=3$) is the unique minimum-cost 3D ground state ($E^*=4.0$, *1:1:1* sphere); `(2,1,1)` ($N=4$) is *4:1:1* prolate ($E^*=4\sqrt{2}$); two `(1,1,1)` triads attract at $d>2R^*$ ($-2m_0/d$) and repel at $d\to 0$ against `(2,2,2)` ($+3.3137$ hard core)
Surrounding-flow Virial lock $T_{ext}/W_{flow}=3$ & integer loop closure $z(1)=z(0)$	Added mass in surrounding flow, quantized line bursts & *180°* annihilation	*75%* of bubble energy in $r\geq R$ flow, *25%* in cavity, *0%* in a point; $(+1,-1)\to 0$ releases two equal *180°* ripples
Coupled finite-parent / receiver continuation & hybrid radial inertia $M(r)=1/(1+r^2/3)$	Wave-driven child opening & last-echo horizon cutoff	Zero-radius opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$; last-echo cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot *25.63%*)
Whole-exterior causal neck onset & harmonic trace $\partial_{\zeta_x} v+\partial_{\zeta_y} v+\partial_{\zeta_z} v=0$	Neck pinch waveform & 3-axis orthogonal cascade $e_z\to e_x\to e_y$	Characteristic cubic $p^3-3p-3=0$ ($p_+=2.1038034$); $1\to 2\to 4\to 8$ lobe cascade with exact harmonic ratios $-1/2$ and -2 and B-independent $\det M_{full}=0.14003084>0$

Strict quarantine of candidate numerical coincidences not derived from A0--A4: To maintain strict separation between proved geometric theorems and post-hoc numerical comparisons, the following expressions are explicitly quarantined from the deductive theorems: (i) Lenz's 1951 number $6\pi^5\approx 1836.1181$ and its dressing $6\pi^5+e^{-3}/\sqrt{2}\approx 1836.1533$ (the discrete **A0+A1** binomial ladder gives $E_{3D}^*(3)/E_{1D}^*(1)=6$ without a derived π^5 continuum factor); (ii) the threshold n–p gap formula $2+(1)/(2)+e^{-3}/\sqrt{2}\approx 2.5352$ m_e; (iii) selecting rung $K=7$ on the η_K ladder without an independent freeze-out equation (especially since Theorem 2 proves three $O(1)$ geometric bypasses of e^{-3K} dilution); (iv) CMB temperature projections using measured H_0 and G or an ad-hoc $-4/5$ exponent; and (v) the single-valve $\Omega_{vis}(\chi_*)=e^{-3}$ ansatz (*71.14%* DE / *23.88%* DM vs. the coupled dynamical visible snapshot *25.63%* in Theorem 4 and the Planck base-ΛCDM fit $68.47\pm 0.73\%$ DE).

Appendix A. Technical atlas and reproducibility index

The main argument follows the outward geometric chain. Sections A.1--A.9 summarize the nine preserved technical engines, and Section A.10 provides the complete reproducibility table binding each section to its verified mathematical owners and numerical datasets. Complete historical chapters, equations, tables, and claim bindings are retained in paper/technical-atlas-record.md and in the archived `v0.26.0` foundation volume.

A.1. Generalized-harmonic evolution engine

The compact nonspherical Einstein/two-scalar generalized-harmonic (GH) implementation, exact frame derivatives, constraint diagnostics and local operator controls evolve the declared covariant action of Section 2. Its existence does not select a cosmic preparation or prove a child; the full technical development is retained in the GH owner.

A.2. Coordinate conditioning

Metric-dependent lapse feedback and proper-duration comparisons control coordinate conditioning without adding physical stress. They remove an imposed coordinate-time ceiling rather than an invariant singularity or a missing vacuum law, as documented in the GH owner.

A.3. Observer readouts

Proper-clock, phase and local stress readouts distinguish lapse/shift coordinates from receiver measurements. Their finite apparatus protocols and supplied initial state are preserved in the GH comparison record without identifying the receiver event with today's cosmological epoch.

A.4. Boson transport

Canonical projection and symplectic matter-probe transport propagate perturbations on the supplied parent geometry. Reference response norms are not energy fractions; incoming covariance and omitted metric fluctuations remain explicit in the boson owner.

A.5. Dense polynomial history

Retained DOP853 dense-polynomial history supplies source-compatible metric and field samples for probe evolution without replacing the nonlinear solution by sparse interpolation or extrapolation, as recorded in the history summary.

A.6. Schur memory kernel

Retarded finite-sector reduction derives the memory kernel from canonical block evolution. Its noise requires an incoming covariance, and initial cross-correlations remain physical data; a memory kernel alone is not a vacuum stress (boson owner).

A.7. Gravitating kick and shear

York-constrained momentum preparations and amplitude-independent shear/internal-coupling relations remain finite conditional results with explicit source profiles and vacuum-reference curvature in the kick owner.

A.8. Finite-amplitude resolution controls

Three-resolution controls at $\varepsilon=0.25$ preserve the nonspherical preparation and fixed proper endpoint with decreasing constraint and adjacent readout differences (comparison record). Separate $\varepsilon=1$ nonlinear runs are **under-resolved, not used for claims** and establish no trapped threshold.

A.9. Matter-work ledgers

Perturbative balances, passive nonlinear ledgers and finite controls retain pressure, geometry, measure and boundary contributions rather than identifying energy loss with vacuum production, with residuals recorded in the coupled-response and GH owners.

A.10. Reproducibility and verification index

All symbolic derivations, numerical solvers, and pre-registered evaluations in Sections 1--9 are executable in src/recursive_horizons/nsc_atlas_offset_generator.py and authenticated against the repository records indexed below:

Manuscript section	Mathematical owner	Authenticated numerical dataset
Section 1 (Offset generator, metric completion, axioms **A0--A4**)	Offset-generator proof	Offset-generator summary
Sections 2--3 (Smooth `0` horizon, flux/area/handoff law)	Parent circulation owner	GH history summary
Sections 4--5 (Pre-registered test NSC-P006, receiver pressure, dust tail)	Offset thermodynamics owner	Flow forecast, Flow evaluation
Section 6 (Finite-amplitude existence, Hebey--Vaugon bound)	Finite-parent data owner	Finite-parent data summary
Section 6.1 (Internal work, continuation, one-mode trapped neck, Dirac mixing)	Valve formation owner	Parent release, Continuation, Surface check, Valve neck, GH neck, Spherical ledger, Conformal continuation, Local restart, Source gate, Dirac modes, Dirac onset, Dirac source

Manuscript section	Mathematical owner	Authenticated numerical dataset
Sections 9.1--9.3 & 9.7 (Theorems 1--3, budget cubic, Friedmann EoS, triad hard core, *3:1* Virial split)	Offset-generator owner	Non-local-to-local atlas summary
Section 9.4 (Theorem 4, coupled parent/receiver, zero-radius opening, last-echo cutoff)	Coupled continuation owner	Finite parent/receiver summary
Section 9.5 (Theorem 5, whole-exterior neck onset $p^3-3p-3=0$, multipole waveform)	Offset-generator owner	Neck waveform summary
Section 9.6 (Theorem 6, 3-axis cascade $e_z \to e_x \to e_y$, 2×2 kinetic handoff)	Offset-generator owner	Inherited daughter summary